

CAPRM-Flood: A Reproducible C++/Python Framework for Property-Level Flood Exposure Indexing

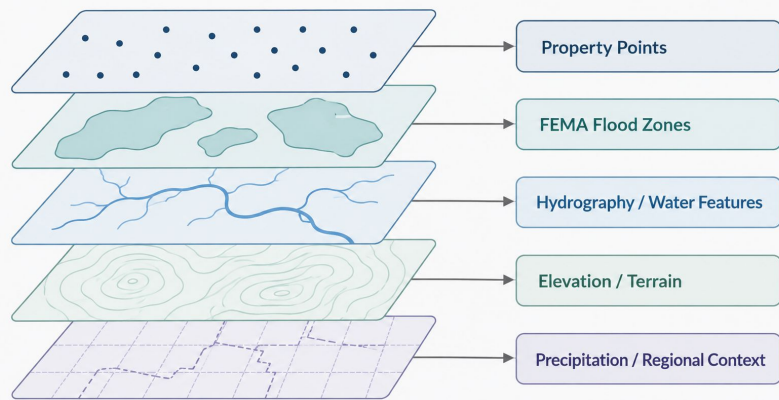
A benchmarked spatial feature-extraction pipeline for property-to-hazard joins

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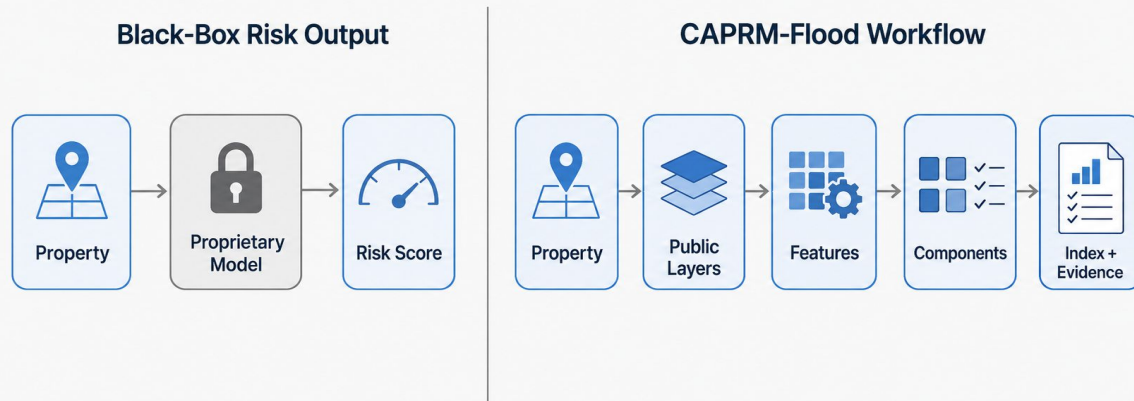
Background: Property-Level Exposure Requires Spatial Joins



- Property-level analysis starts with location points.
- Flood-related evidence exists across separate public geospatial layers.
- Our technical task is to extract consistent features across those layers.
- Correctness, scale, and reproducibility matter.

Motivation: Make the Computation Inspectable

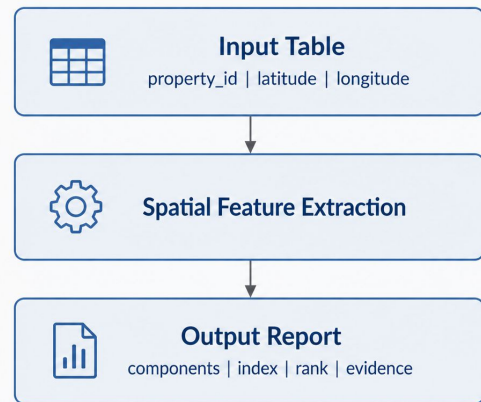
- Some tools emphasize final risk outputs.
- CAPRM-Flood exposes intermediate computation.
- Inputs, features, components, and evidence remain visible.
- The goal is transparent, reproducible exposure indexing.



Problem Statement and Scope

Given property locations and public flood-related geospatial datasets, can we compute transparent property-level exposure features and a relative index efficiently, reproducibly, and with validated spatial operations?

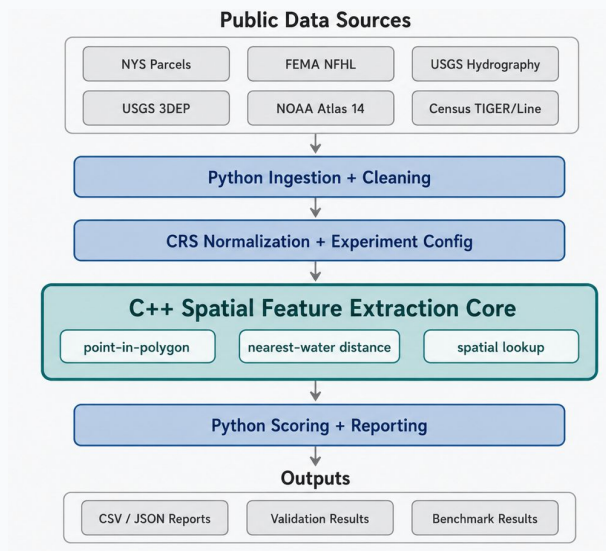
- **Input:** property_id, latitude, longitude
- **Output:** component scores, composite index, percentile rank, evidence fields
- **Scope:** relative exposure within the study region



Solution: Python Pipeline + C++ Spatial Core

- Python handles ingestion, CRS normalization, scoring, reporting, and baselines.
- C++ handles performance-critical batch spatial lookup.
- GeoPandas/Shapely provide correctness baselines.
- Minimum viable target: FEMA point-in-polygon lookup.

- **Input:** property_id, latitude, longitude
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Scoring: Interpretable Components, Explicit Weights

$$\text{Index} = (W_z * C_z) + (W_w * C_w) + (W_e * C_e) + (W_p * C_p) + (W_s * C_s)$$

Feature Source	Extracted Feature	Component
FEMA NFHL	Flood-zone / SFHA membership	C_z
USGS Hydrography	Nearest-water distance	C_w
USGS 3DEP	Elevation / terrain proxy	C_e
NOAA Atlas 14	Precipitation-frequency indicator	C_p
Optional Parcel Attributes	Structure / consequence proxy	C_s

Illustrative output format

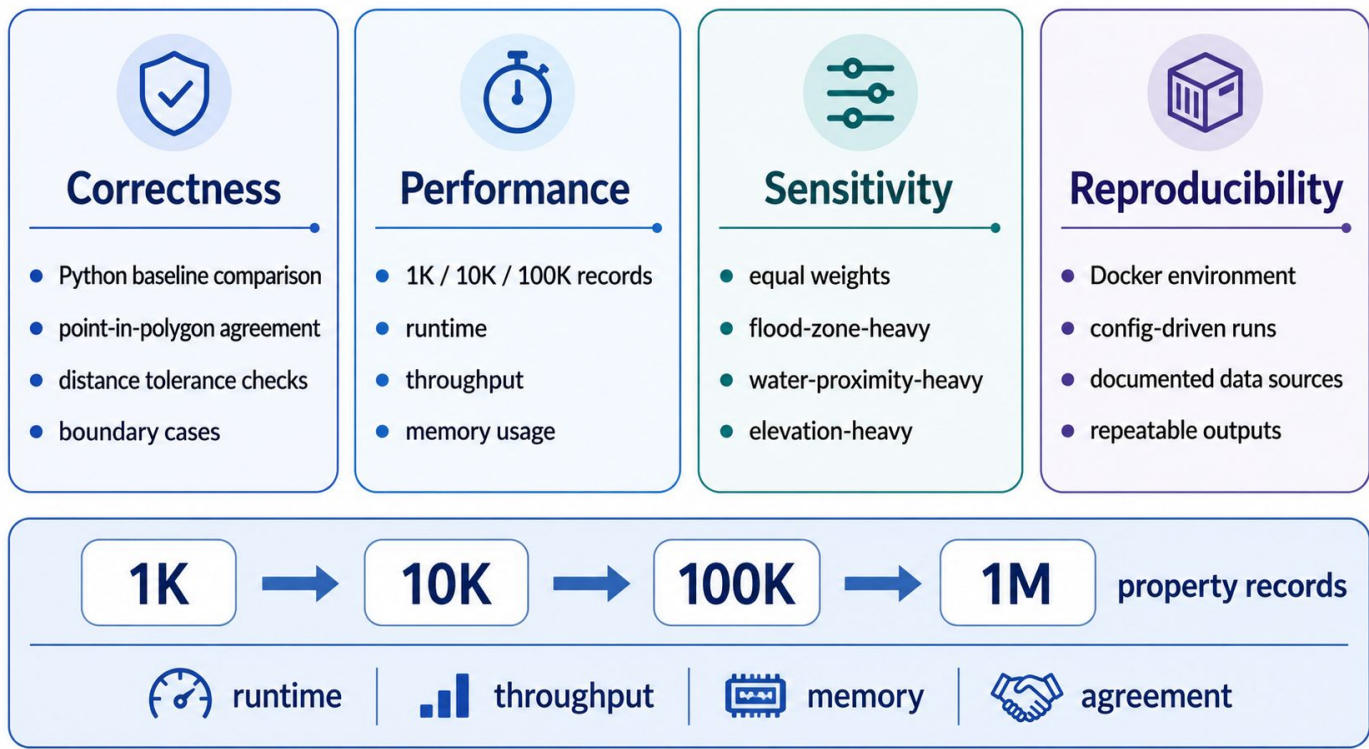
Property 10482

C_z :	90
C_w :	75
C_e :	82
C_p :	70
C_s :	optional

✓ **Composite Index: 81 / 100**

 **Regional Rank: 93rd percentile**

Evaluation Plan: Correctness, Performance, Sensitivity, Reproducibility



References

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